ETH zürich
Institute of Theoretical Computer Science
Prof. Karl Bringmann, Prof. Rasmus Kyng, Prof. Vera Traub
Manuel Christalla, Christoph Grunau, Paul Paschmanns

Algorithms and Probability **Theory Sheet T4** **FS 2026**

The solutions for this sheet must be submitted on Moodle by 23 April, 15:59.

THE FOLLOWING EXERCISE IS WORTH 2 BONUS POINTS.

Exercise T4.1 – *Recursive dice*

Consider the following experiment: We start by rolling a (fair, six sided) die. Let X be the random variable that equals the number the first die shows. Next, we roll a fair X sided die (that is a die with numbers 1 to X on it, each of which is equally likely to appear). Let Y be the random variable that equals the number shown by this second die.

(a) Compute the joint distribution (gemeinsame Dichte) of X and Y.

(b) Compute the marginal distribution (Randdichte) of Y.

(c) What are the expected value and variance of Y?

THE FOLLOWING EXERCISE DOES NOT GIVE BONUS POINTS.

Exercise T4.2 – *Upper Bounds*

We toss a fair coin $n \geq 1$ times. Let X be the number of times the coin shows "heads". Provide upper bounds on $\Pr[X \geq 0.75\,n]$ using the following tools.

(a) Use Markov's inequality.

(b) Use Chebychev's inequality.

(c) Use Chernoff's bound.

Make sure to justify, why the respective inequality can be applied!